Chapter 13 - Binomial distribution

Counting successes in a fixed number of yes/no trials shows up constantly on the WU BBE mathematics paper. A quiz guess, a free throw, a germination check, a first-contact call resolution - each is the same mathematical object once you name the trial, the success label, and the success probability. This chapter develops that object carefully: from a single Bernoulli trial, to a binomial count, to the probability mass function, to means and variances, to English tails such as "at least" and "at most", and finally to the approximations and pooling traps that appear on harder tasks.

The goal is not to memorise a wall of formulas. It is to recognise the setting quickly, write the right probability or sum, and check claims about means, spreads, and ratios without mixing tools.

Learning objectives

- Recognise a Bernoulli trial and the four conditions of a binomial setting, and spot when a story quietly violates one of them.

- Compute

13.1 Bernoulli trials

Start with one experiment that has only two possible results. Call one result success and the other failure. The probability of success is a fixed number p between 0 and 1, and the probability of failure is therefore $q = 1 - p$. That single experiment is a Bernoulli trial.

The word "success" is a modelling label, not a moral judgement. If you are counting defective items, "success" can mean "defective". If you are counting correct answers, "success" means "correct". What matters is that you decide the label once and keep it fixed while you build X.

It helps to think of the trial as an indicator random variable Y: write $Y = 1$ when the trial succeeds and $Y = 0$ when it fails. Then the long-run average of Y is simply p, because a proportion p of independent repetitions show a 1. The variance of a single indicator is $p(1-p)$. That product is small when p is near 0 or near 1 (almost always fail, or almost always succeed - little room to vary) and largest when $p = 1/2$ (maximum coin-flip unpredictability).

$$E[Y] = p, \qquad \mathrm{Var}(Y) = p(1-p).$$

This variance shape is the seed of many exam traps later: a higher success rate does not automatically mean a more variable count.

Example 1. A basketball player makes a free throw with probability $p = 0.80$. One shot is a Bernoulli trial with success = "make". Then $E[Y] = 0.80$ and $\mathrm{Var}(Y) = 0.80 \times 0.20 = 0.16$, so $\mathrm{SD}(Y) = 0.40$.

Example 2. Compare single-trial variances for several p values: $p = 0.10$ gives 0.09; $p = 0.25$ gives 0.1875; $p = 0.50$ gives 0.25; $p = 0.90$ gives 0.09. A guesser near $p = 0.25$ is often *more* variable, trial by trial, than an expert near $p = 0.90$.

Example 3. On a multiple-choice item with four options and one correct answer, random guessing is a Bernoulli trial with $p = 1/4$. Success = "correct"; failure = "wrong". The same structure appears whether the subject is an exam, a seed tray, or a call-centre contact.

13.2 The binomial setting

Now repeat the Bernoulli trial n times and let X be the number of successes among those n trials. When the repetitions satisfy four conditions, X follows a binomial distribution, written

$$X \sim \mathrm{Bin}(n, p).$$

The four conditions are:

1. Fixed n - you decide in advance how many trials you will run (10 questions, 20 items, 15 calls).

2. Two out

If any condition fails, the plain $\mathrm{Bin}(n,p)$ story is the wrong model. Sampling without replacement from a tiny population makes p drift and creates dependence. Gluing together two machines that truly have different defect rates and pretending the combined count is binomial with one pooled p is another common failure mode (see section 13.7).

Before any calculation on an exam, name three things out loud: what is a trial, what counts as success, and what are n and p. Many false statements quietly swap one of these.

Example 1. A quiz has $n = 10$ questions. A student who guesses every question has $p = 0.25$ per question. If guesses are independent, X = number correct satisfies $\mathrm{Bin}(10, 0.25)$. Fixed n, two outcomes, constant p, independence - all present.

Example 2. Drawing 5 cards from a 52-card deck without replacement and counting aces is not binomial with constant p. After each draw the remaining deck changes, so later probabilities depend on earlier outcomes. A "with replacement" story, or an independent-shot story, is needed for $\mathrm{Bin}(n,p)$.

Example 3. An inspector classifies 20 items; each is correctly classified with probability 0.75, independently. Then $X \sim \mathrm{Bin}(20, 0.75)$ with success = "correct classification". If training improves mid-batch so that later items have a higher p, the constant-p condition breaks.

13.3 Probability mass function

For a binomial count $X \sim \mathrm{Bin}(n,p)$, the probability of seeing exactly k successes is

$$P(X = k) = \binom{n}{k} p^{k} (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

Read the formula as a product of two ideas. First, $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ counts how many different sequences of length n contain exactly k successes (and therefore $n-k$ failures). Second, any one specific sequence with k successes and $n-k$ failures has probability $p^{k}(1-p)^{n-k}$ under independence. Multiply the count of sequences by the probability of each sequence.

Three special cases appear so often that they are worth recognising instantly:

When $p < 1/2$ and n is not tiny, p^{n} shrinks extremely fast. Exam claims that a perfect run exceeds a round percentage (20%, 15%, ?) are often deliberate close calls: compute p^{n} carefully instead of guessing from intuition.

Example 1. Three fair coin flips, success = heads, $n = 3$, $p = 1/2$. Exactly two heads:

$$P(X = 2) = \binom{3}{2}\left(\frac{1}{2}\right)^{2}\left(\frac{1}{2}\right)^{1} = 3 \cdot \frac{1}{8} = \frac{3}{8}.$$

The three sequences are HHT, HTH, and THH - each with probability 1/8.

Example 2. Guesser with $X \sim \mathrm{Bin}(10, 0.25)$. A perfect paper is

$$P(X = 10) = (0.25)^{10} \approx 9.54 \times 10^{-7},$$

about 0.000095%, far below a 0.01% cutoff. A statement claiming the perfect-run chance is less than 0.01% is true; a statement claiming it exceeds 0.01% is false.

Example 3. Compute $\binom{10}{3} = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120$. There are 120 ways to choose which three of ten trials succeed; multiply by $p^{3}(1-p)^{7}$ to finish $P(X = 3)$.

13.4 Mean, variance, and standard deviation

A binomial count is the sum of its trials. Write $X = Y_1 + \cdots + Y_n$ where each Y_i is an independent Bernoulli indicator with the same p. Expectation always adds, even without independence, so

$$E[X] = E[Y_1] + \cdots + E[Y_n] = np.$$

Independence lets variances add as well. Each trial contributes variance $p(1-p)$, hence

$$\mathrm{Var}(X) = np(1-p), \qquad \mathrm{SD}(X) = \sqrt{np(1-p)}.$$

These three formulas are the workhorses for "on average", "more variable", and "standard deviation greater than ?" claims. When two sides share the same n, differences and ratios simplify:

- $E[B] - E[A] = n(p_B - p_A)$

- \frac{E[B]

A higher success rate is not a higher variance. An expert with $p = 0.90$ is typically *less* variable than a guesser with $p = 0.25$, because 0.90 is farther from 1/2.

Example 1. On a 10-question quiz, a guesser with $p = 0.25$ has $E[X] = 2.5$. A prepared student with $p = 0.90$ has $E[X] = 9$. The mean gap is $n(0.90 - 0.25) = 6.5$.

Example 2. Same $n = 10$: guesser variance $10 \times 0.25 \times 0.75 = 1.875$; expert variance $10 \times 0.90 \times 0.10 = 0.90$. The claim "the expert has higher variance" is false - the guesser spreads more.

Example 3. An inspector with $n = 20$, $p = 0.75$ has mean $E[X] = 15$. If certification needs at least 18 correct classifications, the inspector's average lies below the bar. That comparison uses only np; no PMF is required.

13.5 Cumulative probabilities and English tails

Exam language almost never stops at "exactly k". The dangerous words are at least, at most, more than, fewer than, and at least one. Each phrase names a different set of mutually exclusive outcomes, and mutually exclusive probabilities add.

"At least k out of n" is the sum of the individual probabilities for $k, k+1, \ldots, n$. A single term $P(X = k)$ undercounts. On true/false items that merely *write* a sum, check three things: the starting index, the ending index n, and the n and p plugged into the PMF - you often do not need the decimal value at all.

Complements save time on long tails:

$$P(X \ge k) = 1 - P(X \le k-1), \qquad P(X \le k) = 1 - P(X \ge k+1).$$

Example 1. Pass if at least 7 out of 10 are correct:

$P(X \ge 7) = \sum_{x=7}^{10} \binom{10}{x} p^{x}(1-p)^{10-x} = P(7)+P(8)+P(9)+P(10).$

If a statement writes exactly this sum with the scenario's n and p, the statement is true as a matter of notation.

Example 2. With $n = 5$ and $p = 0.2$,

$P(X \ge 1) = 1 - (0.8)^{5} = 1 - 0.32768 = 0.67232.$

Adding $P(1)+\cdots+P(5)$ works but is slower; the complement is the intended shortcut.

Example 3. For $n = 10$, "more than 8" means $X \ge 9$ (only 9 or 10), while "at least 8" means $X \ge 8$. One word changes where the sum starts - read the English before the algebra.

13.6 Comparing two binomial sides

Many BBE-style tasks put two people, machines, or batches side by side - same n, different p, or occasionally different n. The claim types repeat:

1. Perfect run - compare p^{n} with a percentage cutoff; compute precisely near round thresholds.

2. Tail wor

A sentence such as "B is at least 300 times as likely as A to pass" means

$\frac{P_B(\text{pass})}{P_A(\text{pass})} \ge 300.$

You need both tails first. A ratio of 282 is not "at least 300", even if it feels close.

Example 1. Free throws, $n = 10$. Rookie $p_A = 0.55$, all-star $p_B = 0.85$. Clutch bonus = at least 8 makes. The all-star's perfect run is $(0.85)^{10} \approx 0.1969$, just under 20%, so "more than a 20% chance of all 10" is false. The mean gap is $10(0.85-0.55) = 3$ exactly. The rookie's SD is larger because 0.55 sits closer to 1/2 than 0.85 does.

Example 2. Suppose $P_A(\text{pass}) \approx 0.35\%$ and $P_B(\text{pass}) \approx 98.72\%$. Then $P_B/P_A \approx 282$. The claim "at least 300 times as likely" is false. Compute the ratio; do not round it up to the exam's round number.

Example 3. A statement says that inspector A's certification probability equals $\sum_{x=18}^{20}\binom{20}{x}(0.75)^{x}(0.25)^{20-x}$. If the scenario really is $n = 20$, $p_A = 0.75$, and "at least 18", the sum matches the English - the item is testing whether you recognise the tail, not whether you can evaluate it by hand.

13.7 When a sum is not binomial

Independent binomial counts combine cleanly only when they share the same success probability. If $X_1 \sim \mathrm{Bin}(n_1, p)$ and $X_2 \sim \mathrm{Bin}(n_2, p)$ are independent, then

$X_1 + X_2 \sim \mathrm{Bin}(n_1 + n_2, p).$

If instead $p_1 \neq p_2$, the total $S = X_1 + X_2$ is still a sum of independent Bernoulli trials, but those trials no longer share one p. The distribution is called Poisson-binomial. Linearity still gives the mean and, with independence, the variance:

$E[S] = n_1 p_1 + n_2 p_2,$

$$\mathrm{Var}(S) = n_1 p_1(1-p_1) + n_2 p_2(1-p_2).$$

People sometimes form a pooled rate $\bar{p} = E[S]/(n_1+n_2)$ and claim $S \sim \mathrm{Bin}(n_1+n_2, \bar{p})$. That fake binomial matches the mean of S, but its variance $(n_1+n_2)\bar{p}(1-\bar{p})$ is generally not equal to $\mathrm{Var}(S)$. Equal means do not make equal distributions.

Example 1. Line A: $n_1 = 10$, $p_1 = 0.9$. Line B: $n_2 = 10$, $p_2 = 0.6$. Independent. Then $E[S] = 15$ and $\bar{p} = 0.75$. True variance: $0.9 + 2.4 = 3.3$. Pooled $\mathrm{Bin}(20, 0.75)$ has variance 3.75. So S is not $\mathrm{Bin}(20, 0.75)$.

Example 2. Two identical machines, each $\mathrm{Bin}(8, 0.7)$ and independent, do combine: the total is $\mathrm{Bin}(16, 0.7)$. Same p is the key.

Example 3. A claim that "the combined defect count follows $\mathrm{Bin}(n_1+n_2, \bar{p})$" is the red flag whenever the story clearly gives two different success probabilities. Trust linearity for the mean; distrust the pooled binomial label.

13.8 Poisson approximation

When n is large, p is small, and the product $\lambda = np$ stays moderate, binomial probabilities sit close to a Poisson law with mean λ:

$$P(X = k) \approx e^{-\lambda}\frac{\lambda^{k}}{k!}.$$

This is the natural approximation for rare events in a long run of trials - defects in a large batch, mutations, occasional server errors - when a task invites the Poisson model. Always check the qualitative conditions: large n, small p, moderate λ. If p is not small, do not lean on Poisson just because λ looks convenient.

Example 1. With $n = 200$ and $p = 0.01$, $\lambda = 2$. Then $P(X = 0) \approx e^{-2} \approx 0.1353$ and $P(X = 1) \approx 2e^{-2} \approx 0.2707$. Exact binomial values are close; the Poisson form is faster by hand.

Example 2. With $n = 10$ and $p = 0.4$, $\lambda = 4$, but p is not small and n is not large. Prefer the exact binomial (or a calculator), not Poisson.

Example 3. If a statement says a rare-event binomial "is approximately Poisson with $\lambda = np$" and the numbers satisfy large n, small p, moderate λ, the approximation claim is in the right regime - still verify the arithmetic of λ itself.

13.9 Normal approximation

For large n with both np and n(1-p) at least about 5, the binomial histogram is roughly bell-shaped and can be approximated by a normal curve with the same mean and variance:

$$X \approx \mathcal{N}\big(np,\, np(1-p)\big).$$

Because X is discrete and the normal is continuous, serious tail work usually needs a continuity correction. For example,

$$P(X \ge k) \approx P\big(Y \ge k - \frac{1}{2}\big)$$

for $Y \sim \mathcal{N}(np, np(1-p))$. Always check the np and n(1-p) rule of thumb on each side before accepting a normal-approximation claim on a true/false item.

Example 1. For $n = 40$, $p = 0.3$: $np = 12$ and $n(1-p) = 28$, both at least 5, so a normal approximation is reasonable. For $n = 20$, $p = 0.05$: $np = 1$ is too small - do not trust the normal rule of thumb (Poisson may fit better if p is tiny).

Example 2. Take $X \sim \mathrm{Bin}(50, 0.4)$ and $P(X \ge 25)$. Mean 20, variance 12, $\mathrm{SD} \approx 3.46$. With continuity correction, approximate by $P(Y \ge 24.5)$ and

$$z = \frac{24.5 - 20}{\sqrt{12}} \approx 1.30,$$

then read a standard normal tail.

Example 3. A false statement often applies the normal approximation when $np < 5$ or $n(1-p) < 5$. Checking those two products is enough to reject the claim without finishing a z-table calculation.

13.10 Skewness (shape)

The binomial distribution is not always symmetric. Its skewness is

$$\gamma_1 = \frac{1 - 2p}{\sqrt{np(1-p)}}.$$

If $p < 1/2$, skewness is positive (long right tail). If $p > 1/2$, skewness is negative (long left tail). If $p = 1/2$, skewness is zero and the distribution is symmetric about its mean. As n grows with p fixed away from the extremes, $|\gamma_1|$ shrinks and the shape becomes more bell-like - which is consistent with the normal approximation.

Example 1. For $n = 20$, $p = 0.2$: $\gamma_1 = 0.6 / \sqrt{3.2} \approx 0.335$ (skewed right). For $p = 0.8$ with the same n, the sign flips and the shape is the left-skewed mirror image.

Example 2. For $n = 20$, $p = 0.5$: $\gamma_1 = 0$, and the distribution is symmetric about the mean 10.

Example 3. Holding $p = 0.2$ fixed while increasing n reduces skewness magnitude, so large-n histograms look closer to a symmetric bell even when p is not 1/2.

13.11 Calculator and exam workflow

On TI-style calculators used in BBE practice:

- binompdf(n, p, k) returns $P(X = k)$
- binomcdf(

Build other tails from complements. In particular,

$$P(X \ge k) = 1 - \texttt{binomcdf}(n,p,k-1).$$

A reliable attack order for true/false statements:

1. Confirm the binomial setting (n, p, independence, two outcomes).
2. Translat

Example 1. For $n = 15$, $p = 0.6$, need $P(X \ge 13)$:

$$P(X \ge 13) = 1 - P(X \le 12) = 1 - \texttt{binomcdf}(15, 0.6, 12).$$

Summing pdf values at 13, 14, and 15 gives the same answer with more keystrokes.

Example 2. Agent A: $n = 15$, $p = 0.60$. Agent B: $p = 0.88$. Bonus if at least 13 resolutions. Perfect run for B: $(0.88)^{15} \approx 0.147$ (under 15%). Mean for B: $13.2 > 13$. SD for A: $\sqrt{15 \times 0.6 \times 0.4} \approx 1.90$ (not greater than 2). One scenario supports several statement types at once.

Example 3. If two statements look similar, differ them by the English: "at least 13" versus "exactly 13", or "more than 25 times as likely" versus "at least 25 times". The algebra follows the words.

13.12 Quick formula sheet

$$P(X = k) = \binom{n}{k} p^{k}(1-p)^{n-k}$$

$$E[X] = np, \qquad \mathrm{Var}(X) = np(1-p), \qquad \mathrm{SD}(X) = \sqrt{np(1-p)}$$

$$P(X \ge 1) = 1 - (1-p)^{n}, \qquad P(X = n) = p^{n}, \qquad P(X = 0) = (1-p)^{n}$$

$$\text{Poisson: } \lambda = np; \qquad \text{Normal: } X \approx \mathcal{N}(np, np(1-p))$$

$$\gamma_1 = \frac{1-2p}{\sqrt{np(1-p)}}$$

If you can state the setting, expand the PMF in words, move between English and sums, and handle means, variances, ratios, and approximations without swapping tools, you are ready for the practice tasks in this chapter.